Essentials of
economics

Krugman • Wells • Olney

chapter 3

Supply and Demand

Prepared by:

Yvonn Quijano

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chapter 3

K-POP Demon Hunters



Fans paid hundreds of dollars to see their concert. What about your favorite K-pop group?

What you will learn in this chapter:

- **competitive market**
- ~~**demand curve**~~ **supply curve**
- The difference between **movements along a curve** and **shifts of a curve**
- **equilibrium price** and **equilibrium quantity**
- In the case of a **shortage** or **surplus**, how price moves the market back to equilibrium

Supply and Demand: A Model of a Competitive Market

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A **competitive market** is a market in which there are many buyers and sellers of the same good or service. So no one can influence on determining the market price.

The **supply and demand model** is a model of how a competitive market works.

Supply and Demand: A Model of a Competitive Market

chapter 3

There are five key elements in the supply and demand model:

- The *demand curve*
- The *supply curve*
- The set of factors that cause the demand curve to shift, and the set of factors that cause the supply curve to shift
- The *equilibrium price*
- The way the equilibrium price changes when the supply or demand curves shift

The Demand Curve

The Demand Schedule and the Demand Curve

A **demand schedule** shows how much of a good or service consumers will want to buy at different prices.

A **demand curve** is a graphical representation of the demand schedule. It shows how much of a good or service consumers want to buy at any given price.

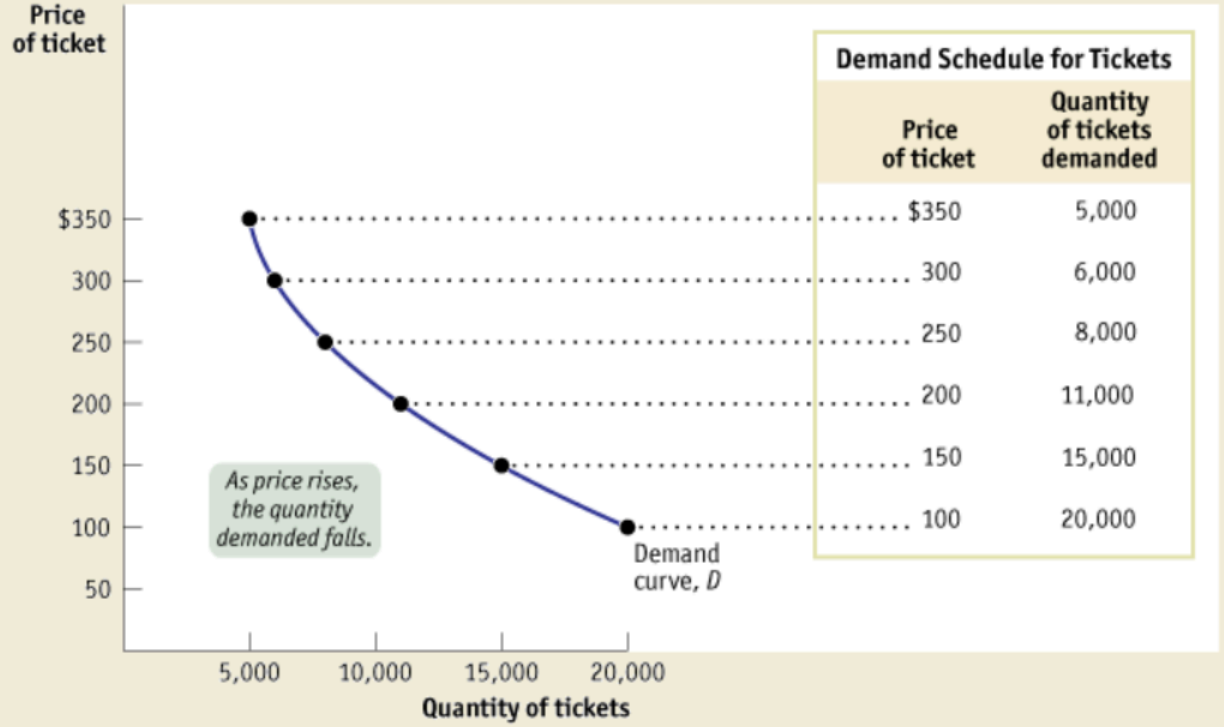
The **quantity demanded** is the actual amount consumers are willing to buy at some specific price.

The Demand Curve

Be careful! In economics, the Y axis is always for the Price, and X axis is the quantity.

The Demand Schedule and the Demand Curve

Figure 3-1 The Demand Schedule and the Demand Curve



Demand Schedule for Tickets

Price of ticket	Quantity of tickets demanded
$350	5,000
300	6,000
250	8,000
200	11,000
150	15,000
100	20,000

The demand schedule for tickets is plotted to yield the corresponding demand curve, which shows how much of a good consumers want to buy at any given price. The demand curve and the demand schedule reflect the law of demand: As price rises, the quantity demanded falls. Similarly, a decrease in price raises the quantity demanded. As a result, the demand curve is downward sloping.

The Demand Curve

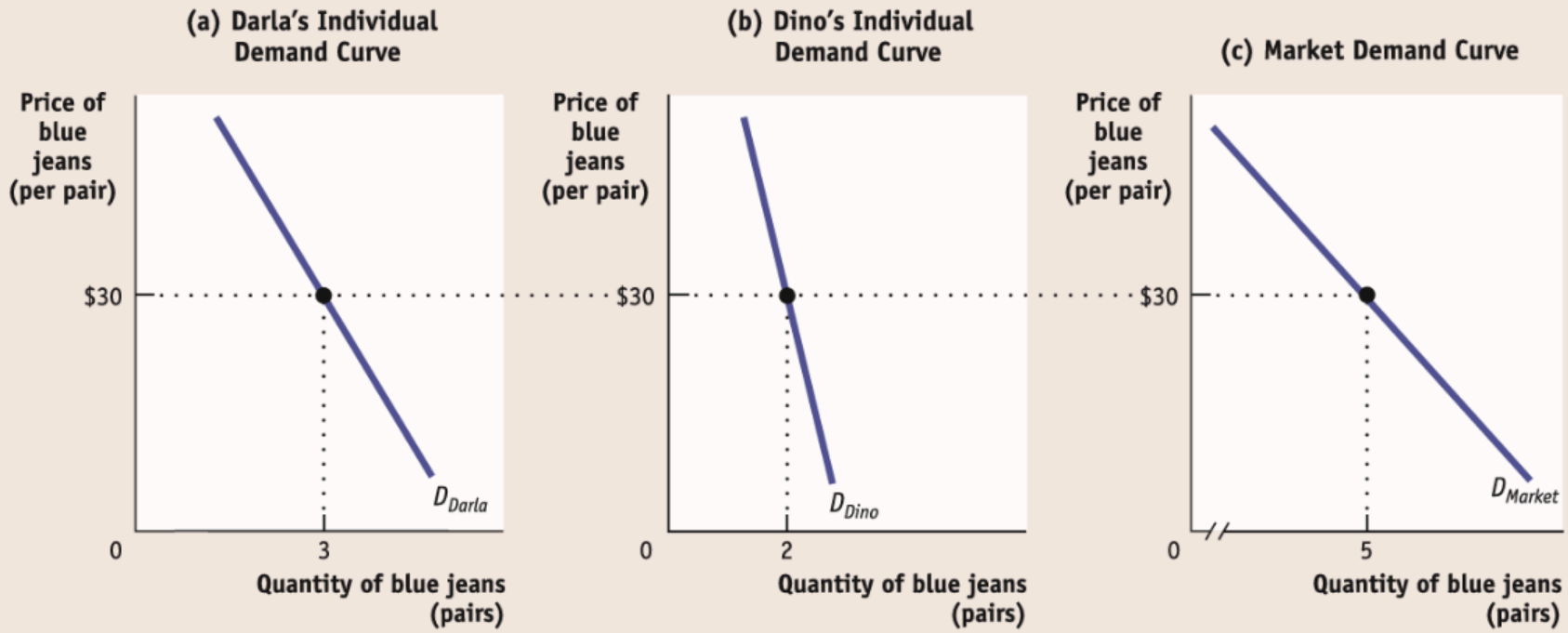
The Demand Schedule and the Demand Curve

The **law of demand** says that a higher price for a good, other things equal, leads people to demand a smaller quantity of the good.

Individual Demand Curves and the Market Demand Curve

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FIGURE 3-5 Individual Demand Curves and the Market Demand Curve

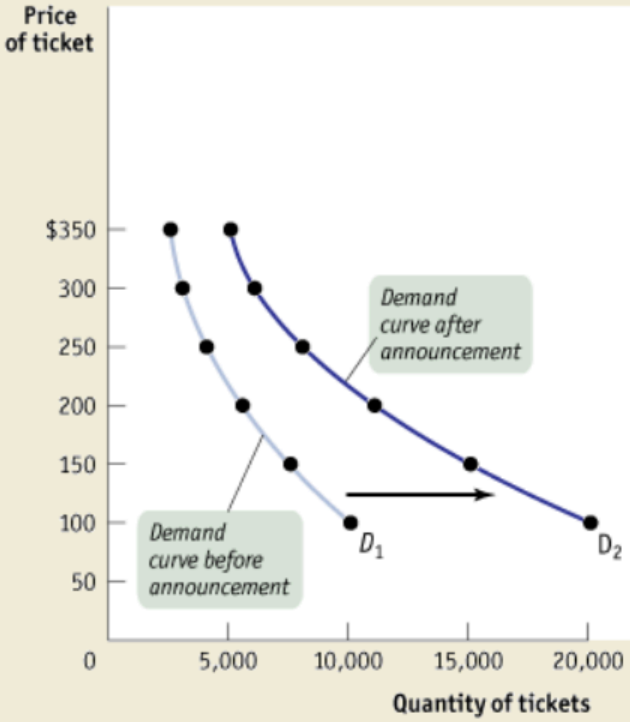


Darla and Dino are the only two consumers of blue jeans in the market. Panel (a) shows Darla's individual demand curve: the number of pairs of blue jeans she will buy per year at any given price. Panel (b) shows Dino's individual demand curve. Given that Darla and Dino are the only two consumers, the *market demand curve*, which shows the quantity of blue jeans demanded by all consumers at any given price, is shown in panel (c). The market demand curve is the *horizontal sum* of the individual demand curves of all consumers. In this case, at any given price, the quantity demanded by the market is the sum of the quantities demanded by Darla and Dino.

The Demand Curve

Shifts of the Demand Curve

Figure 3-2 An Increase in Demand



Demand Schedules for Tickets		
	Quantity of tickets demanded	
Price of ticket	Before announcement	After announcement
$350	2,500	5,000
300	3,000	6,000
250	4,000	8,000
200	5,500	11,000
150	7,500	15,000
100	10,000	20,000

Announcement of Gretzky's retirement generates an increase in demand—a rise in the quantity demanded at any given price. This event is represented by the two demand schedules—one showing demand before the announcement, the other showing demand after the announcement—and their corresponding demand curves. The increase in demand shifts the demand curve to the right.

The Demand Curve

Shifts of the Demand Curve

A **shift of the demand curve** is a change in the quantity demanded at any given price, represented by the change of the original demand curve to a new position, denoted by a new demand curve.

It can happen because of other changes than change in its own price. For example, weather, doctor's new advice, taste change.

Other things being equal or not?

A **movement along the demand curve** is a change in the quantity demanded of a good that is the result of a change in its own price.

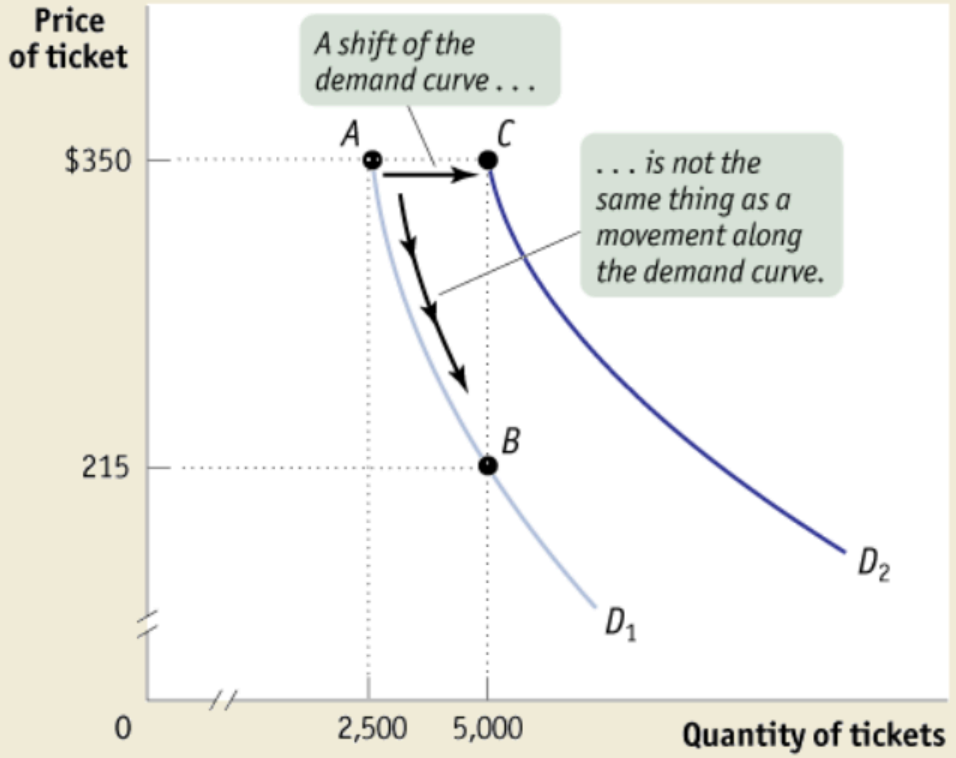
The Demand Curve

Shifts of the Demand Curve

Figure 3-3

Movement Along the Demand Curve Versus Shift of the Demand Curve

The rise in quantity demanded when going from point *A* to point *B* reflects a movement along the demand curve: it is the result of a fall in the price of the good. The rise in quantity demanded when going from point *A* to point *C* reflects a shift of the demand curve: it is the result of a rise in the quantity demanded at any given price.



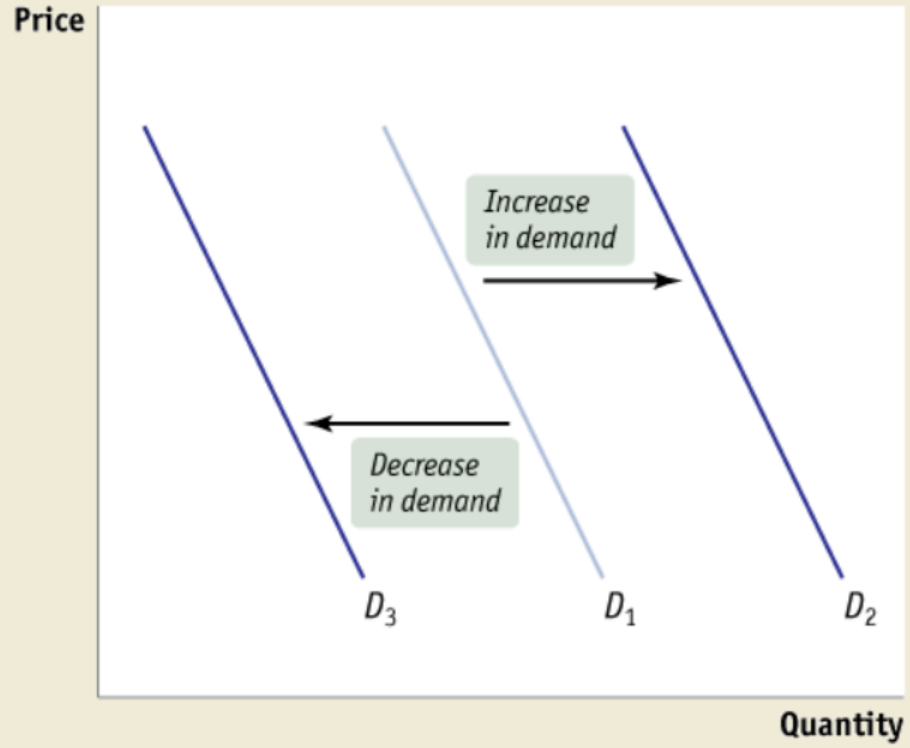
The Demand Curve

Understanding Shifts of the Demand Curve

Figure 3-4

Shifts of the Demand Curve

Any event that increases demand shifts the demand curve to the right, reflecting a rise in the quantity demanded at any given price. Any event that decreases demand shifts the demand curve to the left, reflecting a fall in the quantity demanded at any given price.



The Demand Curve

Understanding Shifts of the Demand Curve

Economists believe that there are four principal factors that shift the demand curve for a good:

- Changes in the prices of related goods
- Changes in income
- Changes in tastes
- Changes in expectations

The Demand Curve

Understanding Shifts of the Demand Curve

Changes in the Prices of Related Goods

Two goods are **substitutes** if a fall (increase) in the price of one of the goods makes consumers less (more) willing to pay the other good: $\frac{\Delta p_X}{\Delta Y} > 0$

ex: Coffee and Tea

Two goods are **complements** if a fall (increase) in the price of one good makes people more (less) willing to pay the other good: : $\frac{\Delta p_X}{\Delta Y} < 0$:

ex: Coffee and Sugar

The Demand Curve

Understanding Shifts of the Demand Curve

Changes in Income

When a rise in income increases the demand for a good —the normal case—we say that the good is a **normal good**.

When a rise in income decreases the demand for a good, it is an **inferior good**.

The Demand Curve

Understanding Shifts of the Demand Curve

Changes in Tastes

People have certain preferences, or tastes, that determine what they choose to consume and these tastes can change. Economists usually lump together changes in demand due to fads, beliefs, cultural shifts, and so on under the heading of changes in *tastes* or *preferences*.

The Demand Curve

Understanding Shifts of the Demand Curve

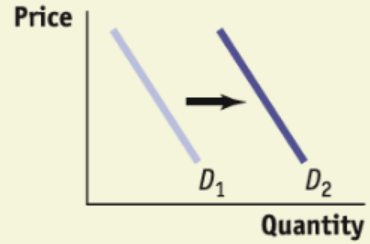
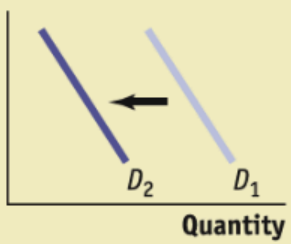
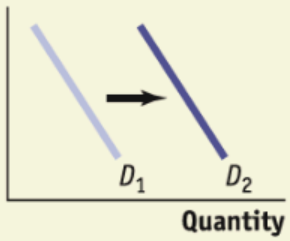
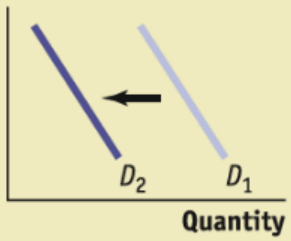
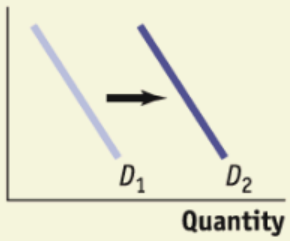
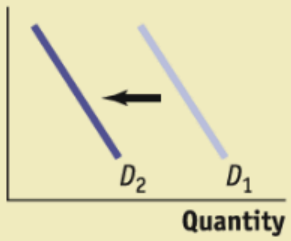
Changes in Expectations

Changes in expectations can either decrease or increase the demand for a good.

Expected changes in future income can also lead to changes in demand.

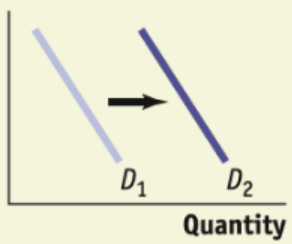
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TABLE 3-1 Factors That Shift Demand

When this happens . . .	. . . demand increases	But when this happens . . .	. . . demand decreases
When the price of a substitute rises . . .	 . . . demand for the original good increases.	When the price of a substitute falls . . .	 . . . demand for the original good decreases.
When the price of a complement falls . . .	 . . . demand for the original good increases.	When the price of a complement rises . . .	 . . . demand for the original good decreases.
When income rises . . .	 . . . demand for a normal good increases.	When income falls . . .	 . . . demand for a normal good decreases.

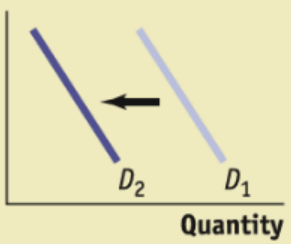
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When income falls . . .



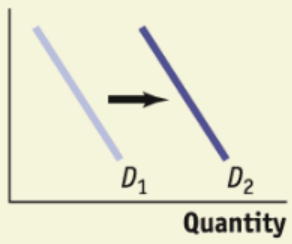
. . . demand for an inferior good increases.

When income rises . . .



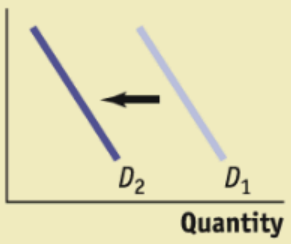
. . . demand for an inferior good decreases.

When tastes change in favor of a good . . .



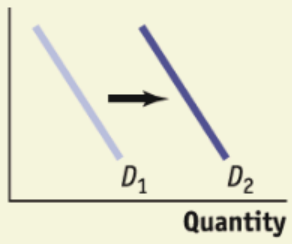
. . . demand for the good increases.

When tastes change against a good . . .



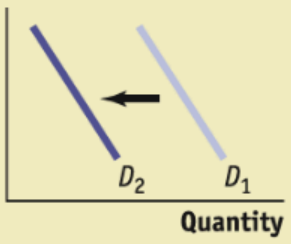
. . . demand for the good decreases.

When the price is expected to rise in the future . . .



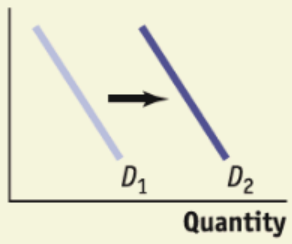
. . . demand for the good increases today.

When the price is expected to fall in the future . . .



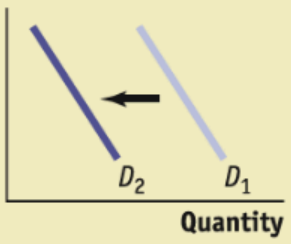
. . . demand for the good decreases today.

When the number of consumers rises . . .



. . . market demand for the good increases.

When the number of consumers falls . . .



. . . market demand for the good decreases.

The Supply Curve

The **quantity supplied** is the actual amount of a good or service people are willing to sell at some specific price.

The Supply Schedule and the Supply Curve

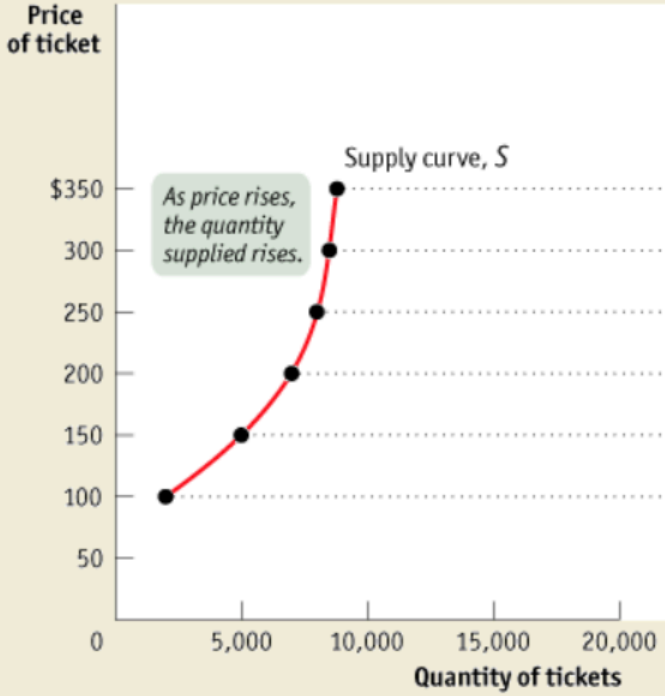
A **supply schedule** shows how much of a good or service would be supplied at different prices.

A **supply curve** shows graphically how much of a good or service people are willing to sell at any given price.

The Supply Curve

The Supply Schedule and the Supply Curve

Figure 3-5 The Supply Schedule and the Supply Curve



Supply Schedule for Tickets

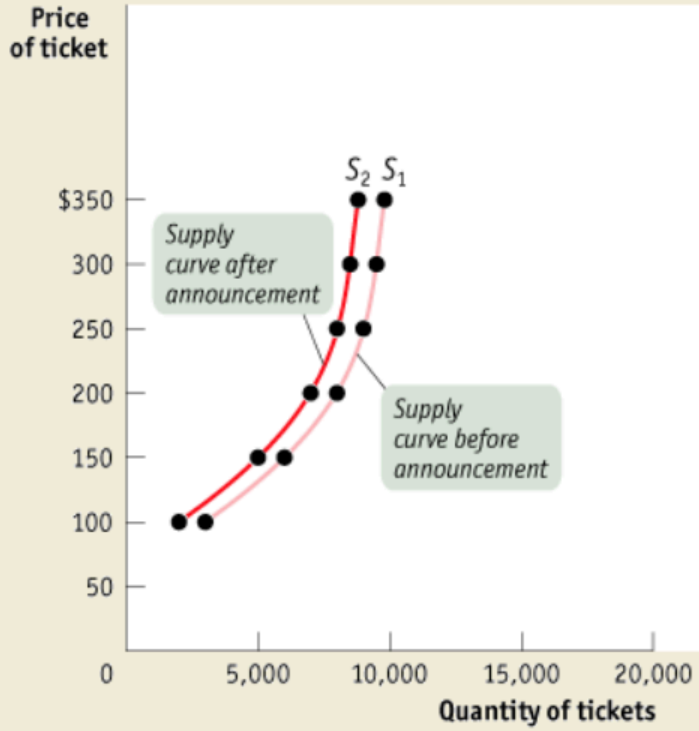
Price of ticket	Quantity of tickets supplied
$350	8,800
300	8,500
250	8,000
200	7,000
150	5,000
100	2,000

The supply schedule for tickets is plotted to yield the corresponding supply curve, which shows how much of a good people are willing to sell at any given price. The supply curve and the supply schedule reflect the fact that supply curves are usually upward sloping: the quantity supplied rises when the price rises.

The Supply Curve

Shifts of the Supply Curve

Figure 3-6 A Decrease in Supply



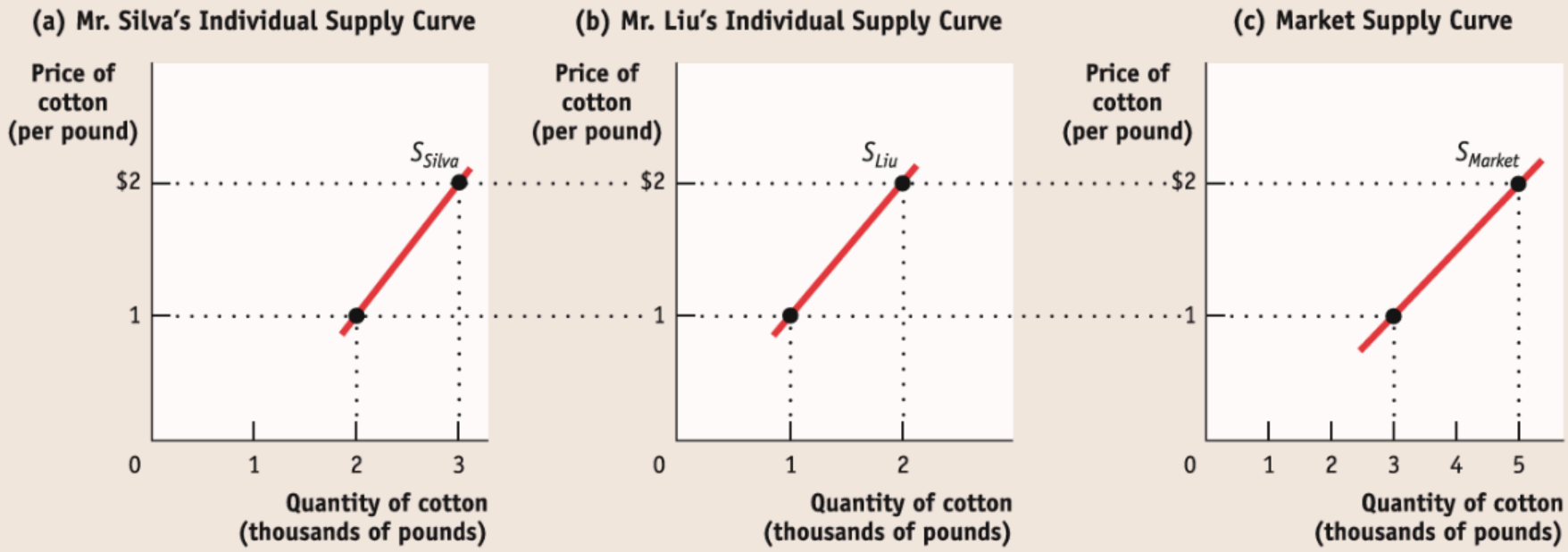
Supply Schedules for Tickets

Price of ticket	Quantity of tickets supplied: Before announcement	Quantity of tickets supplied: After announcement
$350	9,800	8,800
300	9,500	8,500
250	9,000	8,000
200	8,000	7,000
150	6,000	5,000
100	3,000	2,000

Announcement of Gretzky's retirement generates a decrease in supply—a decrease in the quantity supplied at any given price. This event is represented by the two supply schedules—one showing supply before the announcement—the other showing supply after the announcement, and their corresponding supply curves. The decrease in supply shifts the supply curve to the left.

Individual Supply Curve and Market Supply Curve

FIGURE 3-10 The Individual Supply Curve and the Market Supply Curve



Panel (a) shows the individual supply curve for Mr. Silva, S_{Silva}, the quantity of cotton he will sell at any given price. Panel (b) shows the individual supply curve for Mr. Liu, S_{Liu}. The market supply curve, which shows the quantity of cotton supplied by all producers at any given price, is shown in panel (c). The market supply curve is the horizontal sum of the individual supply curves of all producers.

The Supply Curve

Shifts of the Supply Curve

A **shift of the supply curve** is a change in the quantity supplied of a good or service at any given price. It is represented by the change of the original supply curve to a new position, denoted by a new supply curve.

A **movement along the supply curve** is a change in the quantity supplied of a good that is the result of a change in that good's price.

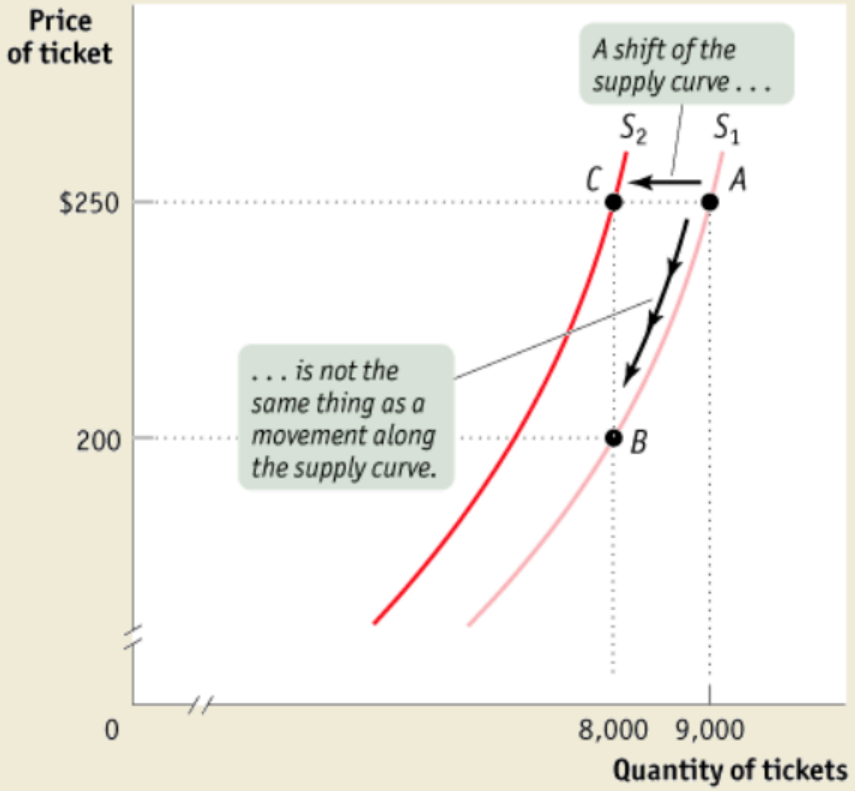
The Supply Curve

Shifts of the Supply Curve

Figure 3-7

Movement Along the Supply Curve Versus Shift of the Supply Curve

The fall in quantity supplied when going from point *A* to point *B* reflects a movement along the supply curve: it is the result of a fall in the price of the good. The fall in quantity supplied when going from point *A* to point *C* reflects a shift of the supply curve: it is the result of a fall in the quantity supplied at any given price.



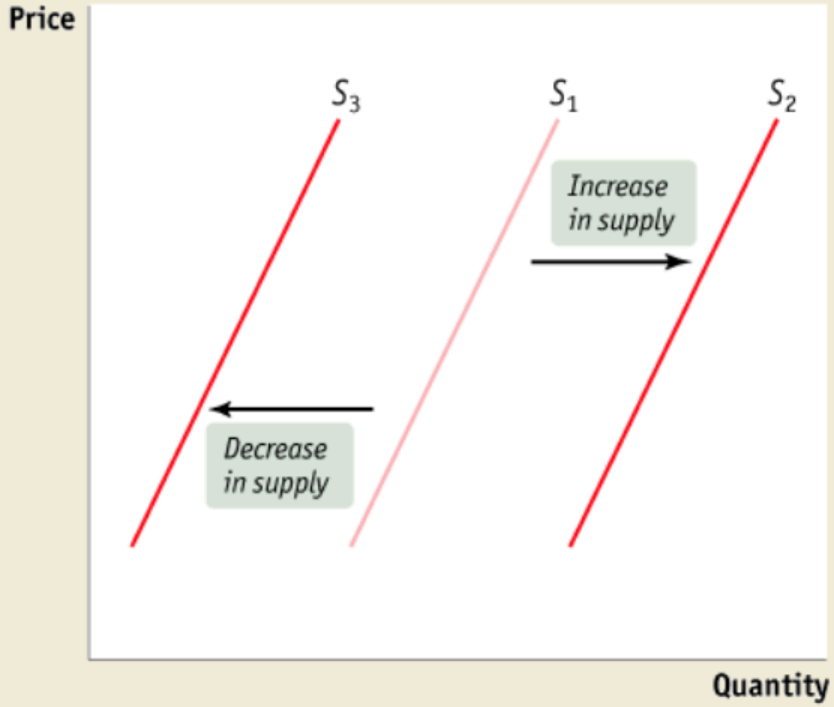
The Supply Curve

Understanding Shifts of the Supply Curve

Figure 3-8

Shifts of the Supply Curve

Any event that increases supply shifts the supply curve to the right, reflecting a rise in the quantity supplied at any given price. Any event that decreases supply shifts the supply curve to the left, reflecting a fall in the quantity supplied at any given price.



The Supply Curve

Understanding Shifts of the Supply Curve

Economists believe that shifts of supply curves are mainly the result of three factors (though, as in the case of demand, there are other possible causes):

- Changes in input prices
- Changes in the prices of related goods or services
- Changes in technology
- Changes in expectations
- Changes in the number of producers

The Supply Curve

Understanding Shifts of the Supply Curve

Changes in Input Prices

An **input** is a good that is used to produce another good.

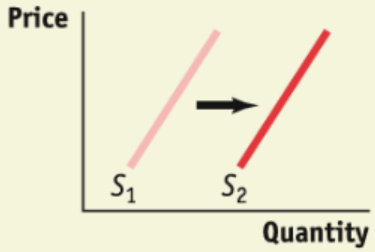
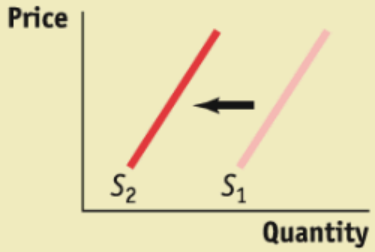
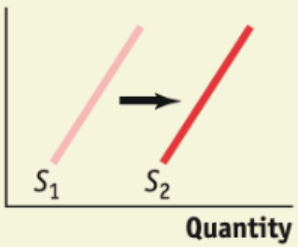
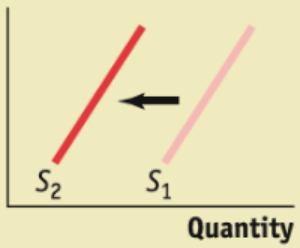
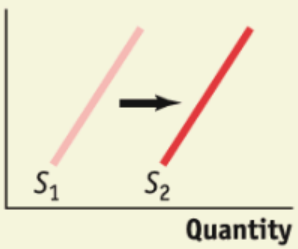
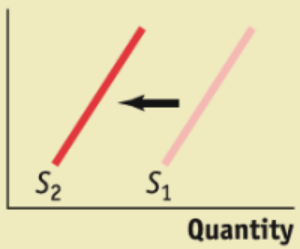
Changes in Technology

When a better technology becomes available that reduces the cost of production, supply increases, and the supply curve shifts to the right.

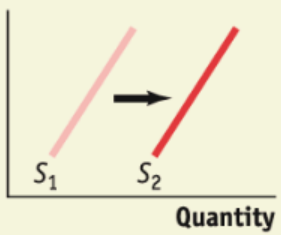
Changes in Expectations

An expectation that the price of a good will be higher in the future causes supply to decrease today, but an expectation that the price of a good will be lower in the future causes supply to increase today.

TABLE 3-2 Factors That Shift Supply

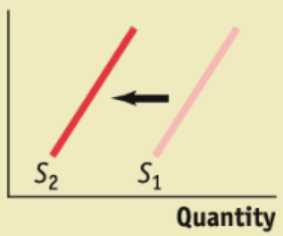
When this happens . . .	. . . supply increases	But when this happens . . .	. . . supply decreases
When the price of an input falls . . .	 . . . supply of the good increases.	When the price of an input rises . . .	 . . . supply of the good decreases.
When the price of a substitute in production falls . . .	 . . . supply of the original good increases.	When the price of a substitute in production rises . . .	 . . . supply of the original good decreases.
When the price of a complement in production rises . . .	 . . . supply of the original good increases.	When the price of a complement in production falls . . .	 . . . supply of the original good decreases.

When the price of a complement in production rises . . .



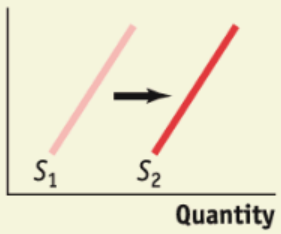
. . . supply of the original good increases.

When the price of a complement in production falls . . .



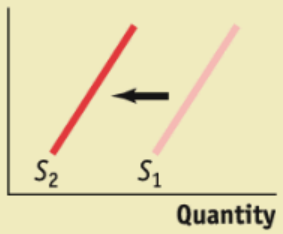
. . . supply of the original good decreases.

When the technology used to produce the good improves . . .



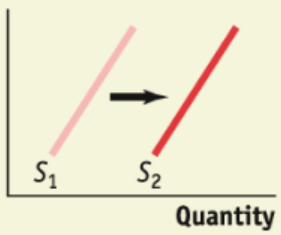
. . . supply of the good increases.

When the best technology used to produce the good is no longer available . . .



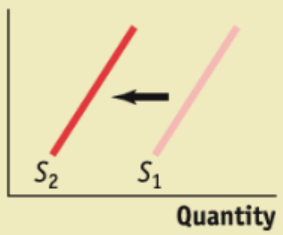
. . . supply of the good decreases.

When the price is expected to fall in the future . . .



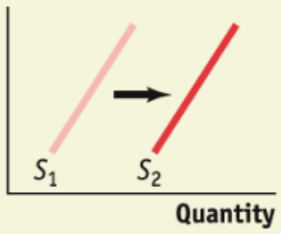
. . . supply of the good increases today.

When the price is expected to rise in the future . . .



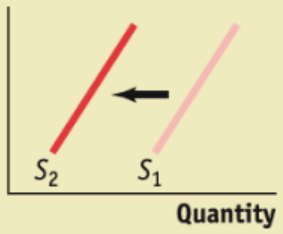
. . . supply of the good decreases today.

When the number of producers rises . . .



. . . market supply of the good increases.

When the number of producers falls . . .



. . . market supply of the good decreases.

Supply, Demand, and Equilibrium

A competitive market is in equilibrium when price has moved to a level at which the quantity demanded of a good equals the quantity supplied of that good. The price at which this takes place is the **equilibrium price**, also referred to as the **market-clearing price**. The quantity of the good bought and sold at that price is the **equilibrium quantity**.

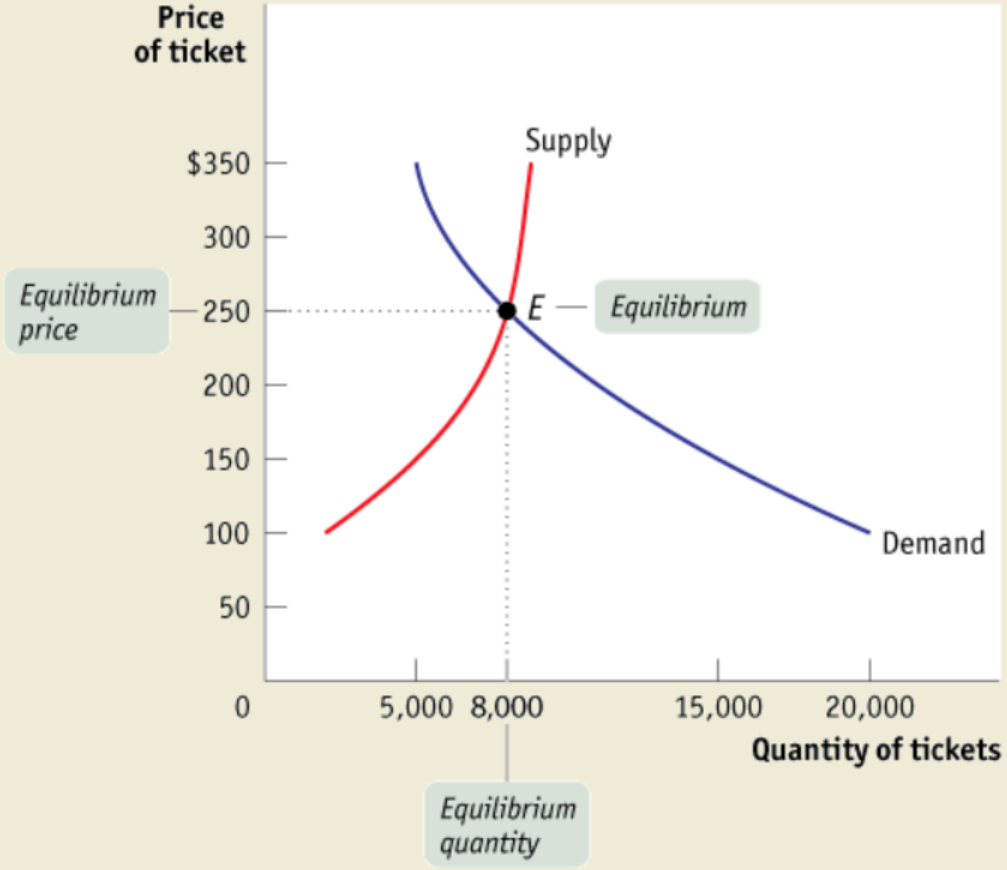
Supply, Demand, and Equilibrium

Finding the Equilibrium Price and Quantity

Figure 3-9

Market Equilibrium

Market equilibrium occurs at point *E*, where the supply curve and the demand curve intersect. In equilibrium, the quantity demanded is equal to the quantity supplied. In this market, the equilibrium price is $250 and the equilibrium quantity is 8,000 tickets.



Supply, Demand, and Equilibrium

Why Do All Sales and Purchases in a Market Take Place at the Same Price?

In any well-established, ongoing market, all sellers receive and all buyers pay approximately the same price. This is what we call the *market price*.



The market price is the same for everyone.

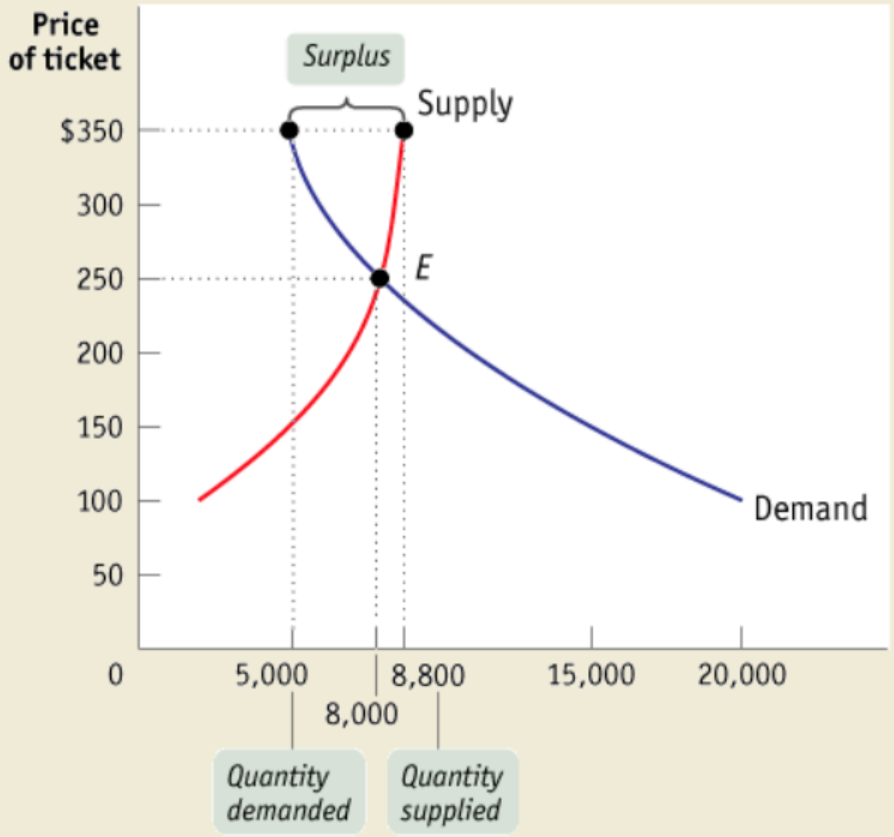
Supply, Demand, and Equilibrium

Why Does the Market Price Fall If It Is Above the Equilibrium Price?

Figure 3-10

Price Above Its Equilibrium Level Creates a Surplus

The market price of $350 is above the equilibrium price of $250. This creates a surplus: at $350 per ticket, suppliers would like to sell 8,800 tickets but fans are willing to purchase only 5,000, so there is a surplus of 3,800 tickets. This surplus will push the price down until it reaches the equilibrium price of $250.



Supply, Demand, and Equilibrium

Why Does the Market Price Fall If It Is Above the Equilibrium Price?

Excess Supply

There is a **surplus** of a good when the quantity supplied exceeds the quantity demanded.
Surpluses occur when the price is above its equilibrium level:

Suppose government has a policy that makes this surplus stays.
What happens? → Buyers market.

Excess Demand

There is a **shortage** of a good when the quantity demanded exceeds the quantity supplied.
Shortages occur when the price is below its equilibrium level.
Suppose government has a policy that makes this shortage stays.
What happens? → Sellers market.

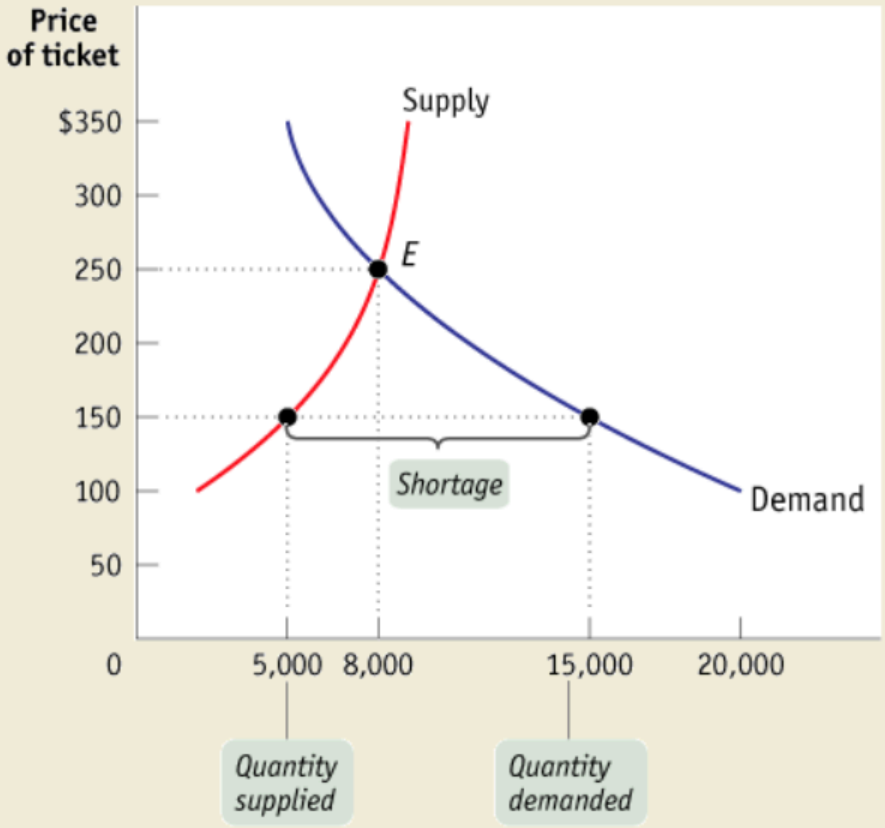
Supply, Demand, and Equilibrium

Why Does the Market Price Rise If It Is Below the Equilibrium Price?

Figure 3-11

Price Below Its Equilibrium Level Creates a Shortage

The market price of $150 is below the equilibrium price of $250. This creates a shortage: fans want to buy 15,000 tickets but only 5,000 are offered for sale, so there is a shortage of 10,000 tickets. This shortage will push the price up until it reaches the equilibrium price of $250.



Supply, Demand, and Equilibrium

Using Equilibrium to Describe Markets

A market tends to have a single price – the market price falls if it is above the equilibrium level but rises if it is below that level.

The market price always moves toward the equilibrium price, the price at which there is neither surplus nor shortage.

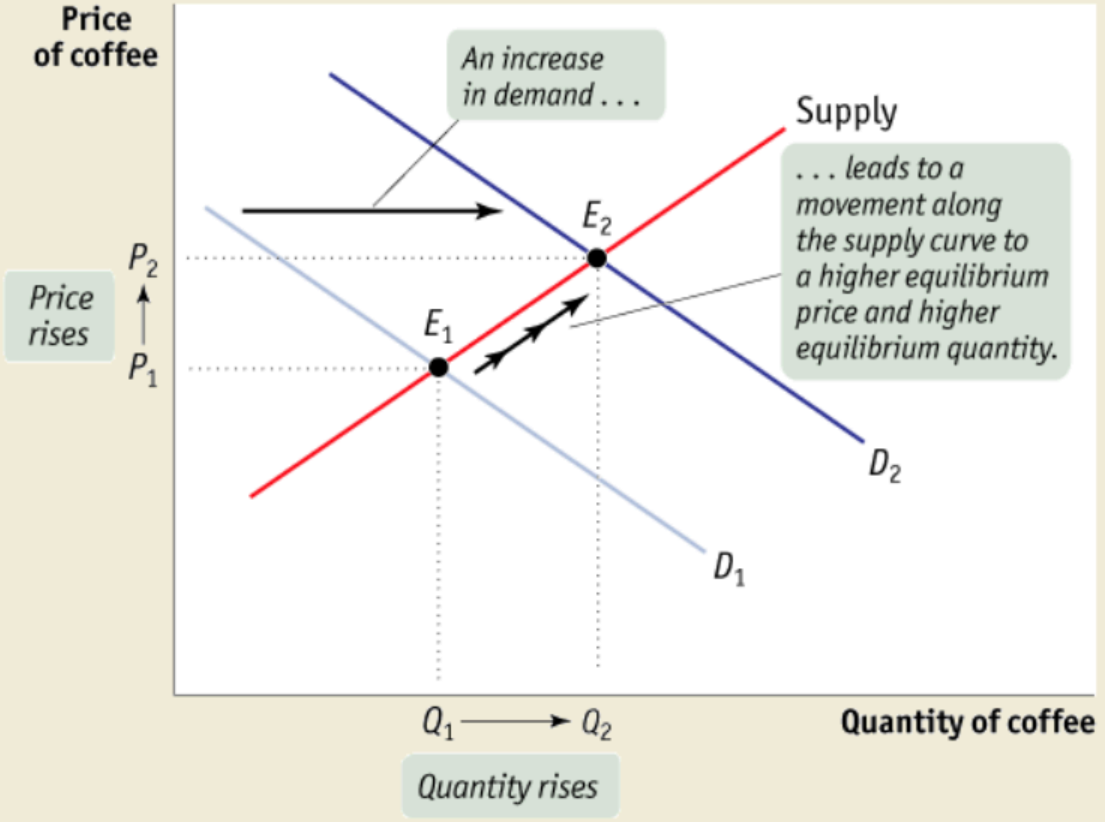
Changes in Supply and Demand

What Happens When the Demand Curve Shifts

Figure 3-12

Equilibrium and Shifts of the Demand Curve

The original equilibrium in the market for coffee is at E_1, at the intersection of the supply curve and the original demand curve D_1. A rise in the price of tea, a substitute, shifts the demand curve rightward to D_2. A shortage exists at the original price P_1, so both price and the quantity supplied rise, a movement along the supply curve. A new equilibrium is reached at E_2, with a higher equilibrium price P_2 and a higher equilibrium quantity Q_2. *When demand for a good increases, the equilibrium price and the equilibrium quantity of the good both rise.*



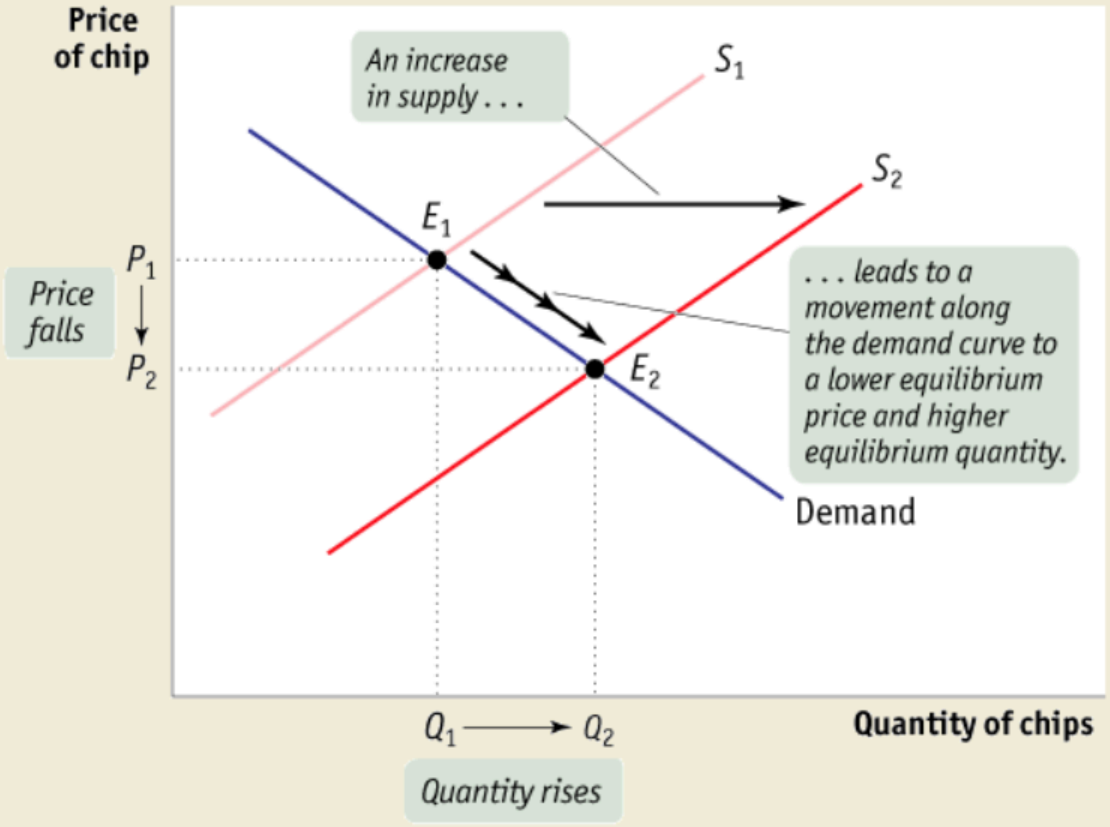
Changes in Supply and Demand

What Happens When the Supply Curve Shifts

Figure 3-13

Equilibrium and Shifts of the Supply Curve

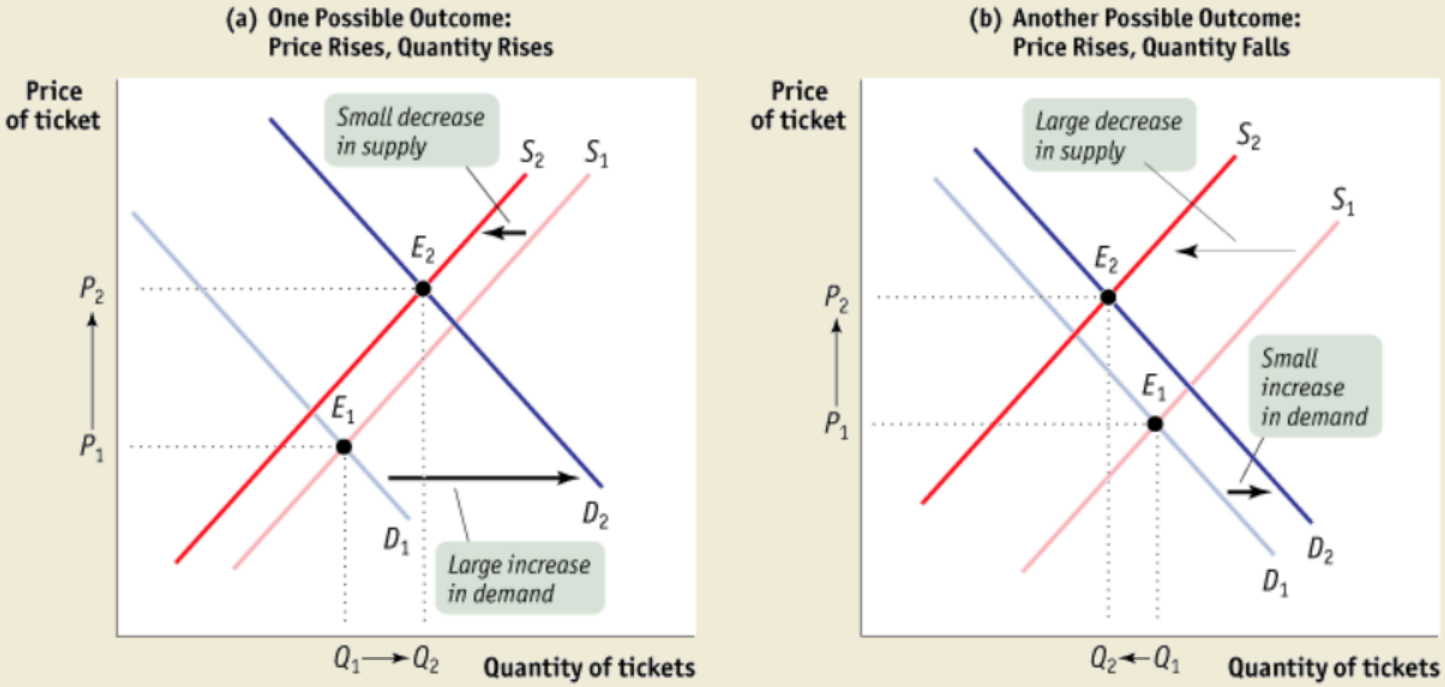
The original equilibrium in the market for silicon chips is at E_1, at the intersection of the demand curve and the original supply curve S_1. After a technological change increases the supply of silicon chips, the supply curve shifts rightward to S_2. A surplus exists at the original price P_1, so price falls and the quantity demanded rises, a movement along the demand curve. A new equilibrium is reached at E_2, with a lower equilibrium price P_2 and a higher equilibrium quantity Q_2. *When supply of a good increases, the equilibrium price of the good falls and the equilibrium quantity rises.*



Changes in Supply and Demand

Simultaneous Shifts in Supply and Demand

Figure 3-14 Simultaneous Shifts of the Demand and Supply Curves



In panel (a) there is a simultaneous rightward shift of the demand curve and leftward shift of the supply curve. Here the increase in demand is relatively larger than the decrease in supply, so the equilibrium price and equilibrium quantity both rise.

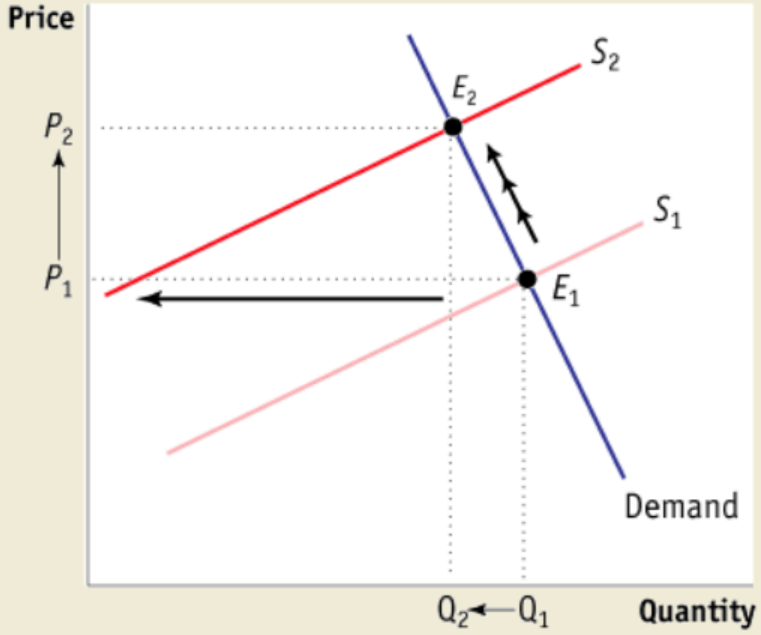
In panel (b) there is also a simultaneous rightward shift of the demand curve and leftward shift of the supply curve. Here the decrease in supply is relatively larger than the increase in demand, so the equilibrium price rises and the equilibrium quantity falls.

FOR INQUIRING MINDS
SUPPLY, DEMAND, AND CONTROLLED SUBSTANCES

Figure 3-15

Effects of the War on Drugs

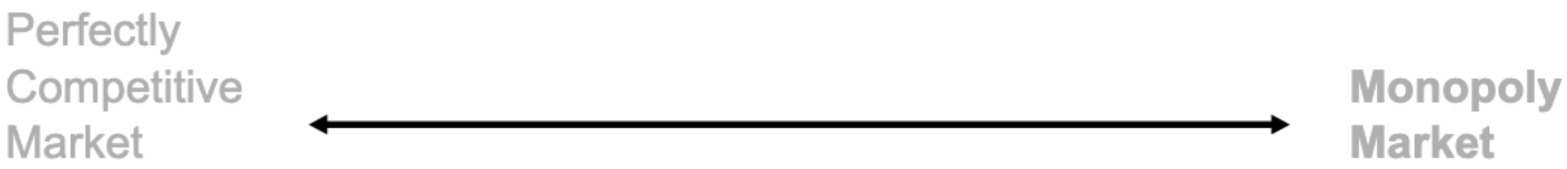
The war on drugs shifts the supply curve to the left. However, we can see by comparing the original equilibrium E_1 with the new equilibrium E_2 that the actual reduction in the quantity of drugs supplied is much smaller than the shift of the supply curve. The equilibrium price rises from P_1 to P_2—a movement along the demand curve. This leads suppliers to provide drugs despite the risks.



chapter 3

Competitive Markets—And Others

When a market is competitive, individuals can base decisions on less complicated analyses than those used in a noncompetitive market. This in turn means that it's easier for economists to build a model of a competitive market than of a noncompetitive market.



KEY TERMS

Competitive market
Supply and demand model
Demand schedule
Demand curve
Quantity demanded
Law of demand
Shift of the demand curve
Movement along the demand curve
Substitutes
Complements
Normal good
Inferior good
Quantity supplied
Supply schedule
Supply curve
Shift of the supply curve
Movement along the supply curve
Input
Equilibrium price
Equilibrium quantity
Market-clearing price
Surplus
Shortage